

Program : Diploma in Information Technology	
Course Code : 3269	Course Title: Python Desktop Application Development Lab
Semester : 3	Credits: 0
Course Category: Program Core	
Periods per week: 4 (L:0 T:0 P:4)	Periods per semester: 60

Course Objectives:

- To provide practical knowledge to develop a simple GUI desktop application using Python.
- To provide very basic knowledge to develop a simple database application.

Course Prerequisites:

Topic	Course Code	Course Title	Semester
Basic IT Skills		Introduction to IT systems	1
Basic programming skill		Problem Solving & Programming	2
Basic programming skill		Problem Solving & Programming Lab	2

Course Outcomes

On completion of the course, students will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Develop Python programs that use control structures and functions.	14	Applying
CO2	Develop Python programs that use Python GUI Library.	15	Applying
CO3	Develop Python programs that use Database.	13	Applying
CO4	Develop Python programs that use Python PDF Library.	15	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Develop Python programs that use control structures and functions		
M1.01	Develop programs using variables and data types.	3	Applying
M1.02	Develop programs using selection and repetition control structures.	3	Applying
M1.03	Develop programs with Lists, Tuples, Sets, Dictionaries	4	Applying
M1.04	Develop programs using functions, class and objects.	4	Applying
CO2	Develop Python programs that use Python GUI Library		
M2.01	Make use of Python GUI widgets like Frame, Label, Entry and Button.	5	Applying
M2.02	Make use of Python GUI widgets like Checkbutton, Radiobutton and ComboBox	5	Applying
M2.03	Make use of Python GUI widgets like Menu, Menubutton.	5	Applying
	Lab Exam – I	1.5	
CO3	Develop Python programs that use Database		
M3.01	Develop Python GUI program to connect a database.	2	Applying
M3.02	Develop Python GUI program to insert values in to a database table.	3	Applying
M3.03	Develop Python GUI program to read values from database table.	3	Applying

M3.04	Develop Python GUI program to update data into a database table.	3	Applying
M3.05	Develop Python GUI program to delete records from a database table.	2	Applying
CO4	Develop Python programs that use Python PDF Library		
M4.01	Develop Python program to create simple report like list of items.	3	Applying
M4.02	Develop Python program to create simple report like list of items using data from database.	3	Applying
M4.03	Make use of Python library to convert report like invoice into PDF.	3	Applying
M4.04	Open-ended experiment. **	6	Applying
	Lab Exam – II	1.5	

**** Suggested Open-ended Experiments:**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open-ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

1. Inventory Management
2. Patient Information
3. Hospital Management

Text / Reference:

T/R	Book Title / Author
T1	Swapnil Saurav, <i>Learn and Practice Python Programming</i> , Eka Publishers, 2020
T2	Alan D. Moore, <i>Python GUI Programming with Tkinter</i> , Packt Publishing, 2018.
T3	Mark Roseman, <i>Modern Tkinter for Busy Python Developers</i> , Late Afternoon Press, 2020

Online Resources:

Sl. No.	Website Link
1	https://realpython.com/python-gui-tkinter/
2	https://www.tutorialspoint.com/python/python_gui_programming.htm
3	https://www.geeksforgeeks.org/python-tkinter-tutorial/
4	https://www.edureka.co/blog/tkinter-tutorial/